



Network Automation Framework

Infrastructure Resiliency & Deployment Orchestration

TechLink Solutions

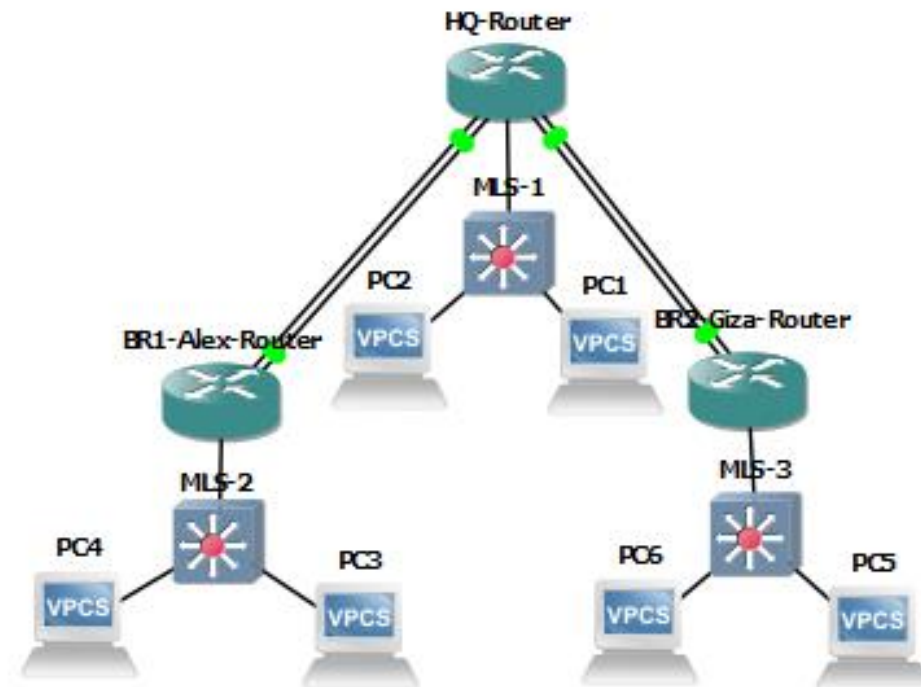
NETWORK TOPOLOGY OVERVIEW

Architecture Design

Strategic multi-branch layout connecting Cairo HQ to Alexa and Giza regional nodes.

- **Dual WAN Ingress:** Redundant ISP paths per branch terminating on Core L3.
- **Path Separation:** HQ Core isolates transit traffic from campus L2 domains.
- **High Availability:** Physical diversity across Gi1/0 to G interfaces.

Logical Segmentation: Strict separation of the management plane (VLAN 10) from end-user traffic (VLAN 20) across all local access ports



IP STRATEGIC ADDRESSING

Segment Role	Prefix	Subnet Mask	Design Rationale
WAN Point-to-Point	192.168.x.x	255.255.255.252 (/30)	Conserve space, minimize broadcast noise
Inter-Site Transit	10.10.1.0	255.255.255.252 (/30)	L3 Core to Distribution Handoff
Branch LAN (Users)	10.11.20.0	255.255.255.0 (/24)	Scalable local user segmentation
Branch Management	10.11.10.0	255.255.255.0 (/24)	Isolated OOB management access

DYNAMIC ROUTING ENGINE



OSPF Area 0

Single Backbone Area 0 configuration for maximum convergence speed and simplicity.



Self-Healing

Sub-second RIB recalculation upon link state failure ensures zero-touch recovery.

SWITCHING & SEGMENT ARCHITECTURE



L2/L3 Access Control

Standardized switching templates across all branches for uniform policy enforcement.

- **VLAN 10:** Management SVI for device telemetry.
- **VLAN 20:** End-user data segment with DHCP relay to Core.
- **Dhcp :** dhcp pool to automate ip distrubition

AUTOMATED PROVISIONING

Configuration Deployment Script (**initial_deploy.py**)

Deployment engine utilizing Python and Netmiko for zero-error baseline configuration.

- Zero-Touch Provisioning" (Just One Click)
- Zero Configuration Drift" (Organized)
- Time/Effort Saved

```
4  # --- CORE ROUTER CONFIG PACKAGES ---
5
6  hq_configs = [
7      'hostname HQ-Router',
8      'interface GigabitEthernet0/0',
9      ' ip address 10.10.1.1 255.255.255.252',
10     ' no shutdown',
11     'interface GigabitEthernet1/0',
12     ' ip address 192.168.12.1 255.255.255.252',
13     ' no shutdown',
14     'interface GigabitEthernet2/0',
15     ' ip address 192.168.13.1 255.255.255.252',
16     ' no shutdown',
17     'interface GigabitEthernet3/0',
18     ' ip address 192.168.22.1 255.255.255.252',
19     ' no shutdown',
20     'interface GigabitEthernet4/0',
21     ' ip address 192.168.23.1 255.255.255.252',
22     ' no shutdown',
23     'interface Loopback0',
24     ' ip address 1.1.1.1 255.255.255.255',
```

LIVE MONITORING TELEMETRY

Telemetry Logic

Continuous route table polling provides human-readable status dashboards every 30 secs.

```
2026-06-21 19:10:09,329 - INFO - -> ALEX Branch Forwarding -> Primary (Gi1/0) IS LIVE
2026-06-21 19:10:09,330 - INFO - -> GIZA Branch Forwarding -> Primary (Gi3/0) IS LIVE

2026-06-21 19:10:39,559 - INFO - [LIVE TELEMETRY TRAFFIC MAP]
2026-06-21 19:10:39,559 - INFO - -> ALEX Branch Forwarding -> Primary (Gi1/0) IS LIVE
2026-06-21 19:10:39,560 - INFO - -> GIZA Branch Forwarding -> Backup (Gi4/0) IS LIVE
```

And When Primary link is down, it's showing that link is up on Backup link immediately

| PHASE 2, What's the Next ??